GRaD-Nav Real4Real audit

Tested dynamics component
Rendering and policy remain blocked

One existing A6000 host. No training. Instance remains running.

Decision: reuse dynamics for research diagnostics. Defer the complete Real4Real workflow.

Real4Real evidence still needed

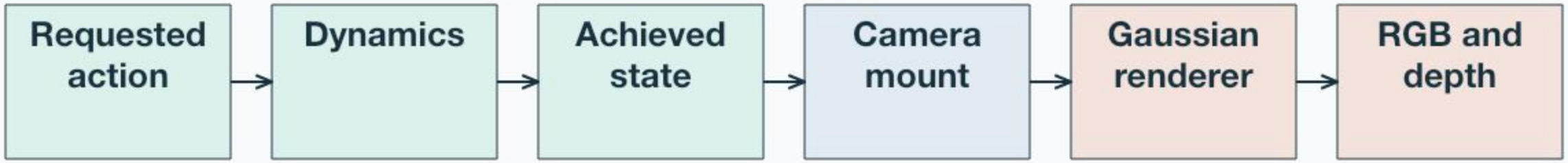
Outdoor residential, suburban and campus scenes below 100 m

Held-out horizontal and vertical camera trajectories

Capture at 15 m and rendering at 50 m remains a research question

Sparse capture coverage and sparse physical content need separate tests

Action, state and observation path



B: measured

Algebra checked

A/D: blocked

C needs the original actor, normalization, VAE and trained visual observation components.

Released scene assets recovered

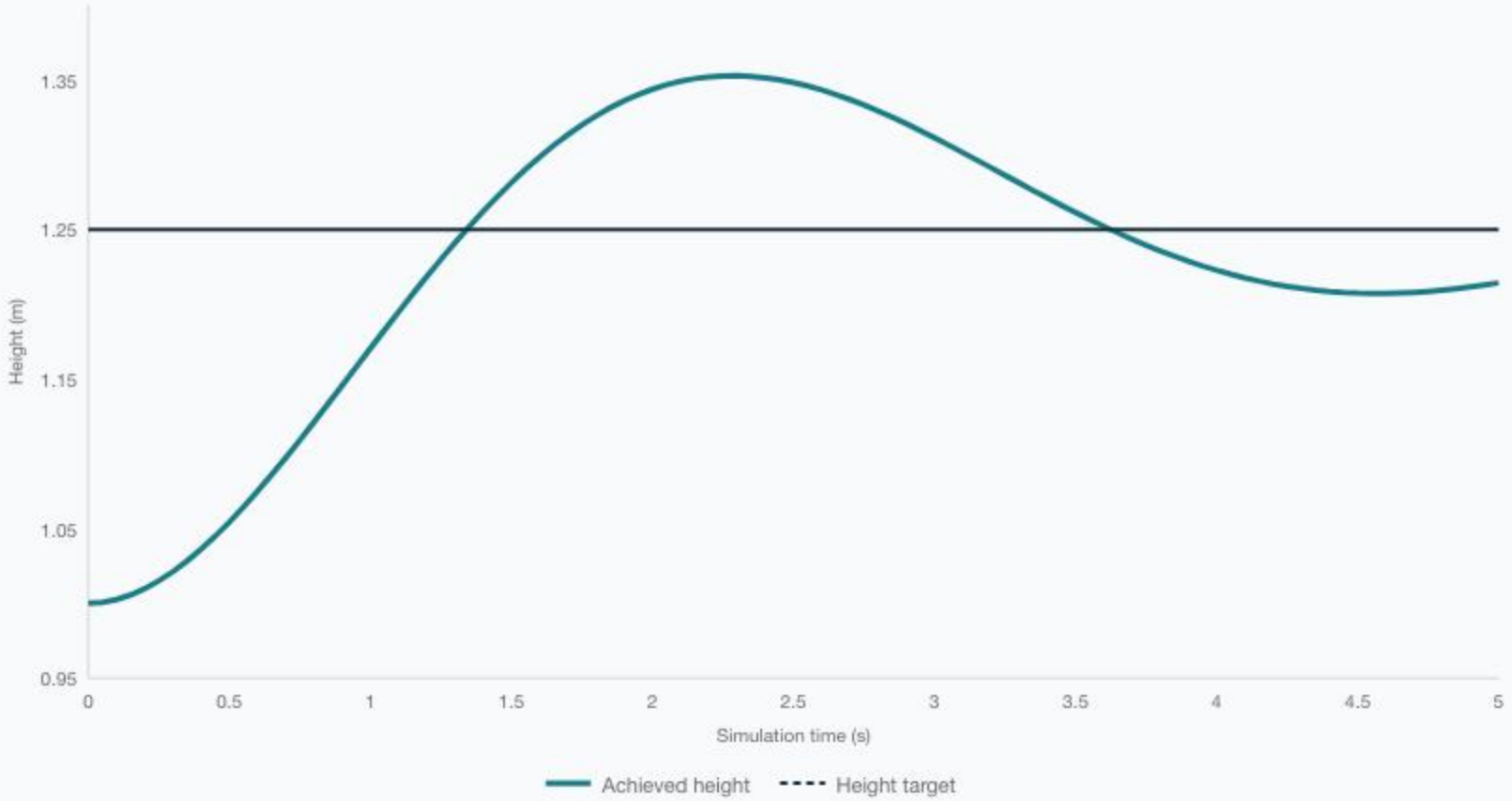
README file IDs returned 404. The author folder supplied replacement IDs.

Gaussian archive 5.049 GB. Full expanded size 5.827 GB.

Point-cloud archive 0.294 GB. Selected cloud 65.9 MB.

Selected gate_mid inputs: 1.140 GB across 1,213 locally verified files.

B feedback response



Final error
0.0356 m

Overshoot
0.1030 m

First 0.02 m band entry
1.25 s

The error band is an authored diagnostic. First entry does not establish settling or stable hover.

B scope and measured behavior

Force balance: 20 × 0.05 s. Height 1.25 m, drift 0.0 at float32 precision.

Feedback: kp 2, kd 0.8. Initial perturbation −0.25 m.

One second: drift +0.1706 m, remaining error 0.0794 m, no overshoot.

Five seconds: drift +0.2144 m, error 0.0356 m, overshoot 0.1030 m.

No saturation. Dynamics median about 14.2 ms, with GPU synchronization. Rendering and policy latency unmeasured.

Numerical camera checks

Camera helper uses xyzw. Dynamics uses wxyz. Conflicting comments are recorded.

Identity, 90° yaw and translation composition checks passed.

Active helper applies no dataparser normalization. Scale/alignment needs scene validation.

Rounded mount orthonormality residual: 0.00042425.

These are algebraic checks. No rendered-camera equivalence or visual calibration was established.

Rendering failure and repair history

gcc: fatal error: cannot execute ‘cc1plus’:
execvp: No such file or directory

nvcc then failed to preprocess host compiler properties.

nvcc 11.8.89. Default gcc 12.3.0. g++ 11.4.0.
cc1plus exists under /usr/lib/gcc/x86_64-linux-gnu/11/.

Two adapter repairs: checkpoint root, scalar camera shape.
Launch correction exposed installed Ninja. A/D remain blocked.

Native policy and collision limits

C blocked: scene checkpoints do not supply the original six learned components.

The native body-rate termination check reads quaternion fields 10:13.

Obstacle handling uses forward-FOV nearest-point distance and reward.

Rear and coincident synthetic points returned infinity. No body-volume contact test.

Outdoor fidelity remains untested

No PSNR, SSIM or LPIPS result. No validated held-out reconstruction split.

No altitude-change, sparse-capture, vegetation or moving-object experiment.

Indoor/gate paper results motivate study. Outdoor and racing were not executed.

Other candidate scenes, controllers and hardware differ. No matched ranking.

Decision, preservation and next evidence

Reuse the tested dynamics component. Defer GRaD-Nav as the complete Real4Real workflow.

All 11 run directories and 1,213 scientific files verified locally.
Logs, configs, route records and compiler failures are preserved.

Later: targeted host-compiler selection, rendered-camera validation, original policy components and trusted held-out real capture.

Provenance appendix

Upstream baseline: d9b00e16107338e49071c65f44582c013e522633

Current source inspected: 95218abe9a5d355519cdaab78d915365303066fe

Host: Ubuntu 22.04.5, RTX A6000, 6 vCPU, 47 GiB RAM

Rendering stack: Torch 2.1.2/cu118, NS 1.1.3, gsplat 1.0.0.

Full-instance charges unknown. $0.54/hour is an unconfirmed GPU-only quote. No termination or cleanup occurred.